

Created environment with osx64 arm (see 01_Deconvolution_scadenCondaEnv.txt file)

```
pip install scaden
```

```
Error: The TensorFlow library was compiled to use AVX instructions,
but these aren't available on your machine.
zsh: abort      scaden process --help
```

<https://github.com/tensorflow/tensorflow/issues/24548>

Remove tensorflow by

```
pip uninstall tensorflow
```

```
pip install tensorflow-cpu==2.13.1 (same version original tensorflow
was).
```

- Uninstalled all tensorflow. Uninstalled scaden.

- Installed scaden with Bioconda

```
pip uninstall charset_normalizer
pip install charset_normalizer==3.3.2
```

- Retried scaden process --help

WORKS

```
2025-05-11 20:41:35.364687: I tensorflow/core/platform/cpu_feature
To enable the following instructions: SSE4.1 SSE4.2, in other op
```



```
Usage: scaden process [OPTIONS] <training dataset to be processed
prediction>
```

Process a dataset for training

Options:

```
--processed_path TEXT  Path of processed file. Must end with .l
--var_cutoff FLOAT      Filter out genes with a variance less than
                        specified cutoff. A low cutoff is recommended
                        should only remove genes that are obviously
                        uninformative.
--help                  Show this message and exit.
```

```
(scadenEnvX86) nv4@mib120460s ~ % scaden --help
```

- Saved a .h5ad object.
- Running scaden process.
- Some commands weren't working, updated by
Pip install scaden==1.1.2
- Conflict with numpy
Pip install numpy==1.22.1
- Scaden simulate to create training (bulk) dataset from snRNAseq reference.
- Do we need to pre-process the data? Had to go back and re-create the SCE because of the duplicated gene names.
 - Option 1: Pre-process using counts but with Scaden .ipynb parameters. Will filter cells and genes.
 - Option 2: Take our log norm counts. I prefer doing this for consistency BUT it differs a little bit from the original counts.
 - Option 3: No normalisation.
- I saved our counts and log norm counts (this would be the output of scaden process on an h5ad obj). Split them into two folders so that the sn data and cell type files have the same prefix.

```
scaden simulate --out ./simulate -d ./scadenInput/countsFinal --
pattern snCounts_counts.txt -p 01_Deconvolution_scaden_snCounts
```

```
TypeError: reorder_categories() got an unexpected keyword argument
'inplace'
```

- Downgraded pandas
Successfully uninstalled pandas-2.0.3
Successfully installed pandas-1.2.4

- WORKED code below!

```
scaden simulate --out ./simulate -d ./scadenInput/countsFinal --
pattern snCounts_counts.txt -p 01_Deconvolution_scaden_snCounts
```

```
scaden simulate --out ./simulate -d ./scadenInput/countsFinalHvgs --
pattern snCountsHvgs_counts.txt -p
01_Deconvolution_scaden_snCountsHvgs
```

```
scaden simulate --out ./simulate -d ./scadenInput/logCountsFinal --
pattern snLogCounts_counts.txt -p
01_Deconvolution_scaden_snLogCounts
```

```
scaden simulate --out ./simulate -d ./scadenInput/logCountsFinalHvgs
--pattern snLogCountsHvgs_counts.txt -p
01_Deconvolution_scaden_snLogCountsHvgs
```

- To run custom scaden (from edited GitHub clone), created conda envX64 and ran pip install -e . From the githubClones/scaden folder. Downgraded pandas and numpy again.

To enable the following instructions: SSE4.1 SSE4.2, in other

operations, rebuild TensorFlow with the appropriate compiler flags.

– Error reading in output files from custom Scaden functions –
> saved as older Ann data.

— Clean environment —

```
create_x86_conda_environment scadenCustomEnvX86 python=3.8
pip install -e ./gitClones/scaden (i.e. not from pip install scaden)
conda list -e > scadenCustomEnvX86Requirements.txt
```

Then ran into error

```
scaden process --help
```

The TensorFlow library was compiled to use AVX instructions, but these aren't available on your machine.

Tried

```
Pip uninstall TensorFlow
```

```
Conda install TensorFlow
```

Tried

```
scaden simulate --help
```

```
pip uninstall charset_normalizer
```

```
pip install charset_normalizer==3.3.2
```

```
pip install numpy==1.22.1
```

```
pip install pandas==1.2.4
```

Tried

```
scaden simulate --help -> works!
```

```
scaden process --help -> works!
```

```
conda list -e > scadenCustomEnvX86Requirements.txt
```

There may still be an issue with how I wrote the files with the custom scaden simulate script (ipynb) because of conflicting anndata version.

– Regenerating .h5ad simulate objects using custom BulkSimulator class.